Vidush Singhal

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Education

- **Purdue University**, West Lafayette, IN, USA
  Ph.D. in Electrical and Computer Engineering — May 2021 – May 2027 (Expected)
- **Purdue University**, West Lafayette, IN, USA
  M.S. in Electrical and Computer Engineering — May 2021 – May 2023
- **Purdue University**, West Lafayette, IN, USA
  B.Sc. in Computer Engineering — Aug 2017 – May 2021

Research Interests

I am a Ph.D. student in Electrical and Computer Engineering at Purdue University working on compiler techniques for improving the performance of programs over irregular recursive datatypes. My thesis is *Compiler Techniques for Optimizing the Layout of Irregular Recursive Datatypes*. My research interests include compilers, programming languages, program analysis, static analysis, data layout optimization, locality optimization, SIMD vectorization, automatic parallelization, and compilers for GPUs. I have experience building compiler infrastructure in Gibbon, LLVM, Clang, and MLIR/Torch-MLIR.

Experience

- **Microsoft Research** [🌐] — *May 2025 – August 2025*
  Research Intern — Redmond, WA, USA
  - Developed **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
  - Implemented a **Clang-to-Clang** source transformer (**~7K LOC of C++**) that emits Pulse code for F* verification.
  - Built a C benchmark suite and validated generated verification conditions on representative annotated C programs.
  - Implemented and verified components of **DPE** (**Dice Protection Environment**) written in C through C2Pulse.
- **Lawrence Livermore National Laboratory** [🌐] — *May 2024 – August 2024*
  Computing Scholar Intern — Livermore, CA, USA
  - Engineered an LLVM Attributor pass to **optimize byte layout within stack allocations** for improved spatial locality.
  - Implemented **LLVM IR** analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
  - Optimized the **LLVM GPU sanitizer** by hoisting redundant checks out of loops, reducing XSBench runtime by **6%**.
  - Built an LLVM IR basic-block permutation pass to evaluate performance sensitivity to code layout.
- **Nod.ai Labs** [🌐] — *May 2022 – August 2022*
  Machine Learning Compiler Engineering Intern — Santa Clara, CA, USA
  - Contributed to **Torch-MLIR**, a compiler stack for lowering programs from the **PyTorch ecosystem to MLIR**.
  - Implemented kernels and compiler support for multiple **PyTorch operators** in the Torch-MLIR pipeline.
  - Added support for the **Facebook DLRM** recommendation model in the PyTorch-to-MLIR compilation pipeline.
  - Investigated improvements to the **ILP solver** used by the Alpa scheduling framework.
- **Astrome Technologies Private Limited** [🌐] — *May 2019 – August 2019*
  Engineering Intern — Bengaluru, Karnataka, India
  - Integrated an eMMC IP block with an AXI-Lite-to-Wishbone bridge using a SystemVerilog wrapper.
  - Verified the eMMC IP and bridge integration through Vivado behavioral simulations.
  - Built XSLT and Python tooling to render XML data as internal HTML reports.
  - Implemented convolutional encoding and Viterbi decoding algorithms in C for error correction and detection.
  - Achieved an **order-of-magnitude speedup** over the original Python implementation.
- **Purdue University** [🏛] — *May 2021 – Present*
  Graduate Research Assistant — West Lafayette, IN, USA
  - **Gibbon Compiler**
    - Optimized layouts of recursive datatypes to improve spatial locality.
    - Measured speedups of **1.14x to 54x** over the best prior work.
    - Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.

    - Structure-of-arrays transformation enables selective traversal, our benchmarks saw speedups of **1.1x to 12x**.
    - Developing optimizations enabled by the SoA representation such as loopification and vectorization.
  - **Orchard Compiler**
    - Developed a compiler framework to automatically fuse and parallelize tree traversals.
    - Measured speedups ranging from **1x to 5x** over baseline tree traversals.
  - **Copse Compiler**
    - Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.
  - **Cornucopia**
    - Developed a fuzzing-based framework to generate a corpus of optimized binaries from source programs using `llc` optimization flags.
    - Discovered bugs in LLVM compiler optimization passes.
    - Exposed decompilation failures in angr, Ghidra, and related binary analysis tools on generated binaries.
    - Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools (BATs).

- **Senior Design Project, SoCET Team [🏛]** *July 2020 – May 2021*
  Senior Design Team Member; advised by Dr. Mark Johnson and Prof. Samuel Midkiff West Lafayette, IN, USA
  - Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
  - Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
  - Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

Publications

C=Conference, J=Journal, T=Thesis

[C.1] Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni *"Vectorized Secure Evaluation of Decision Forests"* in *PLDI 2021* [ACM DL]

[C.2] **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry *"Cornucopia: A Framework for Feedback Guided Generation of Binaries"* in *ASE 2022* [ACM DL]

[C.3] **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni *"Optimizing Layout of Recursive Datatypes with Marmoset"* in *ECOOP 2024* [LIPIcs]

[J.1] **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni *"Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals"* in *TACO 2024* [ACM DL]

[C.4] Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton *"Garbage Collection for Mostly Serialized Heaps"* in *ISMM 2024* [ACM DL]

Coursework

Compilers	Compiler Code Generation and Optimization; Compilers for GPUs; Introduction to Compilers and Translation Engineering
Programming Languages	Reasoning About Programs; Programming Languages; Advanced C Programming; Object-Oriented Programming in C++
Algorithms & Theory	Computation Models and Methods; Applied Algorithms; Data Structures in C; Formal Languages, Computability, and Parallelization
Systems	Computer Architecture; Programming Parallel Machines; Microprocessors and Microcontrollers; Digital Design
Mathematics	Discrete Mathematics; Graph Theory; Category Theory; Random Variables
AI & Data Science	Introduction to Artificial Intelligence; Python for Data Science
Security & Quantum	Holistic Software Security; Introduction to Quantum Computing

Skills

Programming Languages	C; C++; Haskell; Python; Java; Bash; MATLAB; Assembly (ARM, x86); SystemVerilog
Compiler & Verification	LLVM; GCC; Coq; Dafny; Pulse; GDB
Parallel & HPC	CUDA; OpenMP; MPI; Pthreads; OpenCilk; Intel TBB; Intel Cilk Tools
Data & Machine Learning	PyTorch; TensorFlow; NumPy
Systems & Databases	Unix/Linux; Windows; PostgreSQL; AFL++; gem5
Web & Development Tools	HTML; CSS; Git; Visual Studio Code; Vim; CLion; Notepad++; Kate
Professional Skills	Communication; teamwork; inclusive collaboration

Teaching Experience

- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Fall 2025*
- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University *Spring 2023*
- **Undergraduate Teaching Assistant**, Prof. Cheng Kok Koh, ECE 368, Purdue University *Spring 2020*
- **Student Grader for Signals and Systems**, Prof. Chih-Chun Wang, ECE 301, Purdue University *Spring 2020*

Talks Given

- **MWPLS 2025**, AoS To SoA Transformation of packed ADTs *Fall 2025*
- **MWPLS 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **ECOOP 2024**, Optimizing Layout of Recursive Datatypes with Marmoset *Fall 2024*
- **PurPL Seminar**, Optimizing Layout of Recursive Datatypes with marmoset *Spring 2024*
- **ASE 2022**, Cornucopia: A Framework for Feedback Guided Generation of Binaries *Fall 2022*

Honors and Awards

- **Received a travel grant**, LLVM developers meeting *October 2024*
- **Semester Honors**, Purdue University *Fall 2019 - Spring 2020*
- **Deans's List and Semester Honors**, Purdue University *Fall 2017 - Spring 2018*

Leadership Experience

- **Seminar Co-Coordinator**, Purdue Programming Languages and Systems Research Group (PurPL) *Fall 2025 - present*

Review Experience

- **Member of the Artifact Evaluation Committee (AEC)**, ICFP 2026 *Summer 2026*
- **Member of the Program Committe (PC Member)**, C3PO 2026 *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, PLDI 2026 *Spring 2026*
- **Member of the Artifact Evaluation Committee (AEC)**, SPLASH 2026 *Spring 2026*
- **Additional Reviewer**, PPoPP 2026 *Fall 2025*
- **Member of the Reviews for the Journal of Open Source Software**, JOSS *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, POPL 2026 *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, CGO 2026 *Fall 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, ICFP 2025 *Summer 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, PLDI 2025 *Spring 2025*
- **Collaborative Reviewer with Advisor (Milind Kulkarni)**, CGO 2025 *Spring 2025*
- **Member of the Artifact Evaluation Committee (AEC)**, CGO 2025 *Fall 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, PPoPP 2024 *Spring 2024*
- **Member of the Artifact Evaluation Committee (AEC)**, PPoPP 2023 *Spring 2023*

Volunteer Experience

- **Talk moderator**, LLVM developers meeting 2024 *Fall 2024*
- **Student Volunteer**, ECOOP/ISSTA 2024 *Spring 2024*
- **Student Volunteer**, SPLASH 2021 *Fall 2021*

Professional Memberships

- **Eta Kappa Nu**, Purdue Electrical and Computer Engineering Honors Society *Fall 2018 - Present*

Mentoring

- **Logan Anderson**, Phd @ Purdue *Fall 2025*
- **Aniruddh Srivastava**, Undergrad @ Purdue *Fall 2025*
- **Peter A Kaya Gretchikha**, Undergrad @ Purdue *Fall 2024 - Fall 2025*
- **Mikah Kainen**, Undergrad @ Purdue *Fall 2023*
- **Qingyuan (Peter) Li**, Undergrad @ Purdue, SURF summer research mentor *Spring 2023 - Fall 2023*